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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
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TITLE OF THE PROJECT	DataCleanAI: Automated Data Quality, Cleaning & Validation Platform	
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Abstract

Project Background / Problem

Modern organizations collect large volumes of data from databases, spreadsheets, APIs, logs, applications, sensors, and business systems. However, raw datasets frequently contain missing values, duplicate records, inconsistent formats, invalid entries, outliers, incorrect data types, and conflicting values. Poor data quality can lead to inaccurate analytics, unreliable machine-learning models, reporting errors, and inefficient business decisions. Manual data cleaning is often repetitive, time-consuming, and difficult to standardize across different datasets. Users also need a way to validate cleaned data and understand what transformations have been performed. DataCleanAI addresses this problem by providing a centralized platform that automatically profiles datasets, detects quality issues, recommends or applies cleaning operations, validates the resulting data, and produces an understandable quality report.

Objectives

The primary objective is to develop an intelligent platform that can assess the quality of structured datasets before they are used for analytics or machine learning. The system aims to automatically identify missing values, duplicates, inconsistent formats, invalid records, outliers, and schema-related problems. It will recommend suitable cleaning strategies based on detected data characteristics and allow users to review or apply those transformations. Another objective is to validate the cleaned dataset against configurable rules and quality thresholds, measure the improvement in data quality, and maintain a history of profiling, cleaning, and validation operations. The project also aims to provide a scalable and user-friendly environment for repeatable data preparation workflows.

Proposed Methodology / Technologies

The platform will follow a modular architecture with React.js and TypeScript for the frontend dashboard and FastAPI for backend data-processing and API services. PostgreSQL will store dataset metadata, validation rules, processing history, and quality scores. Python-based data-processing libraries such as Pandas and NumPy can be used for profiling and transformation tasks. The system will calculate quality dimensions such as completeness, validity, consistency, uniqueness, and accuracy-related indicators. Gemini API can be integrated to explain detected issues, recommend appropriate cleaning strategies, generate validation rules from natural-language requirements, and summarize data-quality reports. Docker will containerize the application, Kubernetes can manage deployment and scaling, Prometheus and Grafana can support monitoring, and GitHub Actions can automate CI/CD. Trivy can be used to scan container images for security vulnerabilities, while Git and GitHub provide version control and collaboration.

Key Features

The platform will include dataset upload and schema inspection, automatic data profiling, data-quality scoring, missing-value detection, duplicate detection, data-type and format validation, outlier identification, inconsistent-value detection, and rule-based validation. Users will be able to preview problematic records, review recommended cleaning operations, and apply transformations such as missing-value imputation, duplicate removal, standardization, normalization, type conversion, and invalid-record handling. The system will compare quality metrics before and after cleaning and generate downloadable reports containing detected issues, applied transformations, validation results, and final quality scores. An AI assistant can provide explanations for detected anomalies and help users define dataset-specific validation rules.

Expected Outcomes

The expected outcome is a reliable and reusable data-quality platform that reduces the manual effort required for dataset preparation and improves confidence in downstream analytics and machine-learning workflows. The system should provide transparent profiling and validation results so users can understand both the original quality problems and the effects of each cleaning operation. By maintaining processing history and validation results, the platform can support repeatable and auditable data-preparation pipelines. The cloud-native design will demonstrate containerization, scalable deployment, monitoring, CI/CD, and security practices in addition to core data-engineering capabilities. Future extensions can include support for additional file and database sources, automated pipeline scheduling, domain-specific validation templates, drift detection, and advanced AI-assisted data-quality recommendations.

Signature of the Guide HOD